Training a Singing Transcription Model Using Connectionist Temporal Classification Loss and Cross-Entropy Loss

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***Abstract*—In this paper, we propose a method that uses a combination of the Connectionist Temporal Classification (CTC) loss and the cross-entropy loss to train a note-level singing transcription model. By considering the task as predicting a note sequence of the input audio, we can compute the CTC loss between the prediction and the groundtruth note sequence, and further use it with the traditional cross-entropy loss to optimize the transcription model. By comparing the proposed method with a baseline that only utilizes the cross-entropy loss, the results show improved model performance on all the evaluation metrics. Furthermore, using the CTC loss allows the transcription model to learn from weakly labeled data, which is easier to annotate than traditional strongly labeled data. Moreover, we point out the issue of the intrinsic global time shift on the onset labels between datasets. By automatically estimating and calibrating the global time shift of the training dataset, the performance of the singing transcription model is then not affected by the global time shift in the cross-dataset evaluation scenario.**

***Index Terms*—Singing transcription, connectionist temporal classification loss, music information retrieval.**

I. Introduction

NOTE-LEVEL automatic singing transcription is a task that aims to automatically estimate the sequence of sung notes from a music recording that contains a singer's voice. Each note contains the attributes of *onset time*, *offset time* and *note pitch*. The onset time and offset time are not quantized. The *note pitch* refers to the note's pitch value which is quantized to an integer of semitones according to music theory. Such a representation can serve as a high-level feature for other music information processing tasks, such as query-by-humming [1], singing voice separation [2], and singing voice synthesis [3]. A singing transcription system can also be used to generate initial note annotations for singing voice datasets (e.g., used for singing voice synthesis [4]), which greatly reduces annotation time. For applications in the music industry, a singing transcription model can help composers write down sheet music by humming, which has been commercialized by Vochlea.[1] It can also help musicologists efficiently transcribe traditional music pieces which do not have written scores [5].

While the note-level singing transcription task can be considered a special case of music transcription, the diversity of singing voices and styles, such as vibratos and portamenti [6], makes it a unique and challenging task. Traditionally, rule-based algorithms [1], [7] and hidden Markov models (HMM) [8], [9], [10], [11] have been used to transcribe singing voices. Recently, the introduction of deep learning methods [12], [13] and the availability of a large-scale dataset [14] have led to performance improvements.

These recent deep learning methods only utilize the cross-entropy loss for model optimization by encouraging the model to output the same frame labels (onset, offset, and note pitch) [12], [13], [14] as the groundtruth. However, this approach has two shortcomings, which are listed below:

First, the cross-entropy loss computes the loss of each frame separately, which does not match the property of onset and offset prediction of sung notes, and thus is not the best loss function for training note-level singing transcription models. As mentioned in previous work, the onsets and offsets of sung notes are usually smooth [12], which makes their exact positions difficult to label [14] even by expert musicians [6]. On the other hand, even non-expert human labelers can identify the presence of a sung note well [14]. Therefore, instead of only encouraging the model to output the correct frame-level onset and offset labels, it is desirable for a loss function to also encourage the model to predict the correct *note sequence* globally, in which the onset and offset times of the notes are allowed to deviate slightly from the groundtruth. This can be viewed as an easier task that helps the model detect the presence of onsets and offsets of sung notes and avoid note sequence prediction errors that can easily be detected by humans, such as spurious or missed notes. However, the cross-entropy loss can only compute frame-wise losses, but not the loss for the note sequence prediction globally.

Second, strong (dense) labels which contain the precise onset and offset timing labels are required to compute the cross-entropy loss. Therefore, singing data with only weak labels, such as non-aligned musical scores, cannot be used for model training, which restricts the amount of available training data.

Manuscript received 4 March 2022; revised 31 August 2022 and 23 October 2022; accepted 8 November 2022. Date of publication 23 November 2022; date of current version 5 December 2022. The associate editor coordinating the review of this manuscript and approving it for publication was Prof. Hema A Murthy. *(Corresponding author: Jun-You Wang.)*

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Digital Object Identifier 10.1109/TASLP.2022.3224297

[1] [Online]. Available: https://vochlea.com

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To address these shortcomings, in this paper, we introduce connectionist temporal classification (CTC) loss [15] into the model training. CTC loss is a loss function that sums up the probability of all valid alignments that align a model's frame-level outputs with the *target sequence* (the *decoding* process). In the note-level singing transcription task, the *target sequence* is the note sequence. Since the alignments are computed by the loss function itself, the exact alignment (i.e., the precise onset and offset timings) between note labels and the audio is not required. Instead, only the note sequence which represents the order of the presented notes is required.

By using a combination of the cross-entropy loss and the CTC loss, the proposed method does not suffer from the two shortcomings discussed above. For the first issue, since the CTC loss allows all possible alignments of a model's frame-level prediction with the target note sequence, a model that predicts the correct *note sequence* would get a low CTC loss, and vice versa. Therefore, this issue can be resolved. For the second issue, since the CTC loss does not require precise onset and offset timing labels to compute, weakly labeled data can also be used. In practice, we suggest using a small amount of strongly labeled data to compute the cross-entropy loss and a large amount of weakly labeled data to compute the CTC loss.

To verify the effectiveness of the proposed method, we train a singing transcription model to perform note-level singing transcription for *monophonic vocal with instruments*, which is a mixture of monophonic vocals and possibly instruments. The model is trained using different combinations of loss functions. Experimental results show that the combination of the cross-entropy loss and the CTC loss leads to the best model performance. We also test the proposed method in low-resource scenarios, where a small amount of strongly labeled data and a relatively large amount of weakly labeled data is available. In these scenarios, results show that the proposed method achieves an even larger improvement over the baseline. These results are promising, demonstrating the potential applications of the proposed method on more specific transcription tasks (such as traditional or local musical instruments) for which large-scale strongly labeled datasets do not exist.

In addition to the training method, we also point out the intrinsic *global time shift* issue between the groundtruth labels of different singing transcription datasets, which may be due to the different groups of researchers/labelers using different annotation software/interfaces. To address this issue, we propose a method to automatically estimate and calibrate the global time shift between different datasets. By applying this method, the effect of *global time shift* is then alleviated in cross-dataset evaluation.

The rest of this paper is organized as follows: We first briefly introduce previous work in Section II. We then discuss the proposed method and the note-level singing transcription model in Section III. Section IV describes the evaluation metrics and datasets used for model evaluation. The proposed method that estimates the global time shift between different datasets is also discussed in this section. Experimental results are shown in Section V. Section VI gives conclusions and future work.

II. Related Work

We divide related work into two parts, note-level singing transcription systems/datasets, and loss functions.

A. Note-Level Singing Transcription Systems/Datasets

Previous note-level singing transcription systems and datasets can be classified into three categories based on the format of the input signal, namely *monophonic vocal*, *monophonic vocal with instruments*, and *polyphonic vocal*.

1) Monophonic Vocal: Monophonic vocal music consists of only vocals, with at most one note sung at a time. Traditionally, pitch tracking algorithms and hand-crafted note segmentation algorithms have been used to construct transcription systems [1], [7]. These systems first use a hand-crafted note segmentation algorithm to segment sung notes, and then use the results of a pitch tracking algorithm to determine the pitch of each note. Recently, deep neural networks (DNN) [12] have been used as an alternative to hand-crafted note segmentation algorithms. Hidden Markov models (HMM) [8], [9], [10] have also been used for singing transcription, which directly model note sequences. By learning from strongly labeled data, these systems achieve better performance than traditional rule-based systems [8], [10], [12].

To train and evaluate monophonic singing transcription systems, the ISMIR2014 dataset [6] and TONAS dataset [5] have been widely used in the literature [5], [8], [10], [12]. These datasets are labeled and verified by experts, but their scales are not large enough (containing only 2000 to 3000 notes) to allow researchers to train deeper and more expressive models [14].

2) Monophonic Vocal With Instruments: Monophonic vocal with instruments is a mixture of vocals and possibly other instruments, where the vocal part is restricted to be monophonic. For this kind of input, an early work proposed by Ryynänen and Klapuri [11] applied an HMM to model the distribution of sung notes, and used a key estimation algorithm to detect the musical key, which is then used to determine the note transition probability.

In the last few years, with the advancement of singing voice separation (SVS) systems, several studies have adopted the SVS model as a pre-processor to extract vocals from the mixture. For example, Wang and Jang [14] and Omnizart package [13] use Spleeter [16] to extract vocals. Their proposed neural networks then take the extracted vocals as input to perform singing transcription.

In this paper, to validate the effectiveness of the proposed training method, we train a model to transcribe sung notes from *monophonic vocal with instruments* audio. We use the MIR-ST500 dataset [14] for model training and evaluation, which contains 500 pop songs and their note labels. While *RWC Music Database: Popular Music* [17] is also a *monophonic vocal with instruments* dataset that contains note labels, its labels are aligned and quantized at the beat level, and are therefore not suitable for evaluating *note-level* singing transcription systems. Instead, it is more suitable for evaluating *notation-level* transcription systems, which output musical scores with quantized notes, as classified by [18].

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3) Polyphonic Vocal: Singing transcription of *polyphonic vocal* is the most generalized case. There are no restrictions on the input. The groundtruth may contain concurrent notes, which increases the difficulty of building transcription systems. Due to the lack of large-scale datasets and the difficulty of transcribing polyphonic vocal music [19], previous work has only focused on frame-level transcription [20], [21] instead of note-level transcription. The Choral singing dataset [22], Dagstuhl ChoirSet [23], and The Bach Chorales[2] are datasets for polyphonic vocal transcription, which contain the note-level annotation of *a cappella* songs.

B. Loss Functions

Cross-entropy loss has been used to optimize neural networks for note-level singing transcription in previous work [12], [13], [14]. By minimizing the cross-entropy between the prediction and the groundtruth labels, a neural network learns to predict the onset and offset frames [12] as well as note pitches [14].

In addition to the cross-entropy loss, we also use the CTC loss in our work. CTC loss was first proposed by Graves et al. [15]. Given an unaligned target sequence, which represents the order in which groundtruth labels are presented, and the model's predictions, it uses dynamic programming to efficiently compute the probability that the predictions align with the target sequence. Therefore, frame-by-frame labels are not required. By using the CTC loss, a neural network can be trained directly on the training data with only the unaligned groundtruth labels. Due to this property, it has been used in the task of automatic speech recognition (ASR), resulting in performance improvements [24].

In the research field of music information retrieval (MIR), the CTC loss has also been used in score–audio music retrieval [25], music transcription of non-vocal instruments [18], [26], [27], multi-pitch estimation [28], lyrics alignment [29], and the training of pitch-class representation [30], allowing models to be trained on weakly labeled data and achieve competitive performance compared to models trained on strongly labeled data, which cannot be achieved by using only the cross-entropy loss.

Inspired by the success of the CTC loss in various tasks, in this paper, we adopt the CTC loss to train a note-level singing transcription model. However, CTC loss was not designed to guide a model to predict the exact timing of notes (but predict the correct sequence of notes), using only CTC loss may not always help a model predict exact onset and offset timings correctly, which is a key requirement of note-level singing transcription. Therefore, in this paper, we propose a slightly different method that uses the combination of the cross-entropy loss and the CTC loss to train a singing transcription model. More details of these statements will be further discussed in Section III-D and verified in Section V-B.

III. Proposed Method

In this section, we first define the problem formulation of *note-level singing transcription for monophonic vocal with instruments* (hereinafter abbreviated as *singing transcription*). We then discuss the two loss functions used in this work, namely the cross-entropy loss and the CTC loss, and how we employ these loss functions to optimize a singing transcription model. Finally, we introduce the architecture of the proposed singing transcription system.

A. Problem Formulation

Given the input music signal $\mathbf{x}$, a singing transcription system outputs a note sequence $\mathbf{n} := (n^1, n^2, \ldots, n^S) \in \mathbb{R}^{S\times 3}$ which represents the notes sung by the vocalist, where S is the number of notes. Each note is represented by three attributes, *onset*, *offset*, and *note pitch*. Notes in $\mathbf{n}$ are ordered by their onset times and do not overlap each other.

In this paper, we divide the task of singing transcription into three steps, namely *pre-processing*, *frame-level transcription*, and *post-processing*, which are described as follows:

Pre-processing: The pre-processing step converts $\mathbf{x}$ into a feature map of $X \in \mathbb{R}^{C\times T\times N}$, where C is the channel number of the feature map, T is the total number of frames, and N is the number of features per frame in each channel. In practice, X is obtained using constant-Q transform (CQT) [31], C is set to 6, N is set to 384, and T depends on the duration of the input audio. The details of the implementation will be further discussed in Section III-E.

Frame-level transcription: The frame-level transcription step takes the feature map of $\mathbf{X} \in \mathbb{R}^{C\times T\times N}$ as input, and outputs the framewise probability prediction of $\mathbf{y} := (y^1, y^2, \ldots, y^T) \in \mathbb{R}^{T\times M}$, where M is the dimension of the output for each frame, which is set to 18 in practice. These dimensions represent the probability that each frame t:

- Is an onset frame, denoted as $p_{\mathrm{o}}(t)$.
- Is a silence frame, denoted as $p_{\mathrm{s}}(t)$.
- Belongs to a note whose octave number is c, denoted as $p_c(t)$, $c \in \{2, 3, 4, 5\}$.
- Belongs to a note whose pitch class name is pn, denoted as $p_{pn}(t)$, $pn \in \{\mathrm{C}, \mathrm{C\sharp}, \mathrm{D}, \ldots, \mathrm{A\sharp}, \mathrm{B}\}$.

Post-processing: The post-processing step converts the framewise probability prediction $\mathbf{y}$ into the note sequence $\mathbf{n}$, which is the output of the entire singing transcription system.

For the proposed singing transcription system, the *frame-level transcription* step is performed by a neural network, which is trained using the proposed method. For simplicity, we refer to this neural network as the *singing transcription model* in the remainder of the paper.

B. Loss Function

In this subsection, we briefly introduce the two loss functions used in this work, namely the cross-entropy loss and the CTC loss.

1) Cross-Entropy Loss: Cross-entropy loss computes the cross-entropy between two different distributions. For two discrete probability distributions p and q, the cross-entropy of p from q is:

$$CE(q, p) = -\sum_{x\in\mathcal{X}} q(x)\log(p(x)) \tag{1}$$

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where $\mathcal{X}$ denotes the set of possible values such that either $p(x)$ or $q(x)$ is not zero. In the multi-class classification task with only one positive class, q is the groundtruth distribution, which is defined as a one-hot vector. That is, if the groundtruth class label is j, then only $q(j) = 1$, and all other $q(x) = 0$. p is the probability distribution predicted by the model, which has been normalized to 1 beforehand. In this case, the cross-entropy loss can be simplified to:

$$L_{\text{CE}} = -\log(p(j)) \tag{2}$$

This case is also referred to as the *sparse cross-entropy loss*.

2) Connectionist Temporal Classification Loss: Connectionist temporal classification (CTC) loss was first proposed by Graves et al. [15]. It sums up a model's predicted probability of every possible path π that can be aligned with the target (groundtruth) sequence $\mathbf{z}$ by the mapping function $\mathcal{B}(\cdot)$. $\mathcal{B}(\cdot)$ first removes the consecutive repeated labels from the input, then removes all blank symbols, which are special symbols that denote *no symbol* is predicted. In other words, the inverse of the mapping function, $\mathcal{B}^{-1}(\cdot)$, can be viewed as an alignment function that aligns the target sequence with the frame-level predictions. For a classification task, suppose the input feature sequence is $\mathbf{x}$, which has a length of T, and the probability that the model outputs for each class i at the frame t is $p_i(t)$, and the t-th element of the path π is π_t, then, we define the probability of each path $p(\pi|\mathbf{x})$ as:

$$p(\pi|\mathbf{x}) = \prod_{t=1}^{T} p_{\pi_t}(t) \tag{3}$$

Then, the CTC loss is defined as the negative log-likelihood of $p(\mathbf{z}|\mathbf{x})$, which is the sum of all valid π that can be aligned with $\mathbf{z}$. That is:

$$p(\mathbf{z}|\mathbf{x}) = \sum_{\pi \in \mathcal{B}^{-1}(\mathbf{z})} \left(\prod_{t=1}^{T} p_{\pi_t}(t) \right) \tag{4}$$

$$L_{\text{CTC}} = -\,log\left(p\left(\mathbf{z}|\mathbf{x}\right)\right) \tag{5}$$

C. Apply Loss Functions for Model Optimization

In this subsection, we discuss the details of using the two loss functions to optimize the singing transcription model.

1) Cross-Entropy Loss: The output of the singing transcription model includes onset, silence, octave number, and pitch class name prediction for each frame, which can be considered as four separate subtasks. We define the groundtruth labels of these subtasks for each frame as follows:

- $p_{\text{o}}(t) = 1$ for *onset* frames, $p_{\text{o}}(t) = 0$ otherwise.
  A frame is an *onset* frame if its distance to the nearest onset (of any groundtruth note) is less than a threshold θ_{d}. In this work, two different thresholds are tested, which will be discussed in Section V.
- $p_{\text{s}}(t) = 1$ for *silence* frames, $p_{\text{s}}(t) = 0$ otherwise.
  A frame is an *silence* frame if the frame is not covered by any groundtruth note.
- $p_c(t) = 1$ if the note that covers the frame t has the note pitch in octave c, otherwise, $p_c(t) = 0$. If t is a silence frame, then all $p_c(t) = 0$.
- $p_{pn}(t) = 1$ if the note that covers the frame t has the note pitch with pitch class name of pn, otherwise, $p_{pn}(t) = 0$. If t is a silence frame, then all $p_{pn}(t) = 0$.

With these definitions, the prediction of $p_{\text{o}}(t)$ and $p_{\text{s}}(t)$ can be viewed as a classification problem, which can be optimized by the cross-entropy loss. For $p_c(t)$ and $p_{pn}(t)$, by adding another class $p_{\text{N}}(t)$ that represents silence frames, and assigns the label of $p_{\text{N}}(t) = 1$ to silence frames, they can also be viewed as a classification problem, and be optimized by the cross-entropy loss.

In practice, the cross-entropy loss of the whole model is the sum of the cross-entropy loss of the four subtasks.

2) CTC Loss: Different from the cross-entropy loss, which computes the loss of each frame t independently, the CTC loss takes into account the model's predictions for the entire input audio and computes the probability that the model's output aligns with the target sequence $\mathbf{z}$.

To compute the CTC loss, we first define the list of all possible symbols, or *alphabets*, $\mathbb{A}$, as:

$$\mathbb{A} := \{\text{blank}, \text{on}, \text{sil}, 36, 37, \ldots, 83\}$$

where blank denotes the blank symbol, which means that no symbol is outputted in this frame; on denotes the onset symbol; sil denotes the silence symbol. The remaining 48 symbols, from 36 (C2) to 83 (B5), represent that a note with the note pitch of the corresponding MIDI number is present. The total number of distinct symbols is 51.

Target sequence: Then, we convert the note labels $\mathbf{n}$ into a target sequence $\mathbf{z}$ that contains only the symbols in $\mathbb{A}$. For a note sequence of $\mathbf{n} := (n^1, n^2, \ldots, n^S)$, where the note pitch of the ith note n^i is denoted by n^i_{p}, the target sequence $\mathbf{z}$ for computing the CTC loss is defined as follows:

$$\mathbf{z} := (\text{sil}, \text{on}, n^1_{\text{p}}, \text{sil}, \text{on}, n^2_{\text{p}}, \text{sil}, \ldots, \text{on}, n^S_{\text{p}}, \text{sil})$$

Since the target sequence contains onset, silence, and note pitch symbols, the output of onset, silence, and pitch value (octave and pitch class name) probabilities are all involved in the computation of the CTC loss. This is different from the cross-entropy loss, where the four predictions are treated as separated subtasks.

Notably, the use of the blank symbol, which represents *no symbol*, provides flexibility to the singing transcription model. It allows the model not to output a meaningful symbol at every frame. Otherwise, for every frame that is neither an onset frame nor a silence frame, it will be forced to output a note pitch, which could be difficult when the frame is close to the note boundary. For such a frame, the note transition would increase the difficulty of the model prediction.

Posterior probability of each symbol: Then, we define the posterior probability of each symbol predicted by the model. To compute the CTC loss, the sum of the posterior probabilities of all possible symbols should be normalized to 1. Therefore, two additional dimensions are added and served as normalization

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factors, including $p_{\mathrm{b}}(t)$ and $p_{\mathrm{h}}(t)$. $p_{\mathrm{b}}(t)$ represents the probability of blank symbol, while $p_{\mathrm{h}}(t)$ stands for the probability of "have pitch," which is treated as *the sum of all note pitch symbols' probability*. A softmax function is applied on the outputs for normalization, making $p_{\mathrm{o}}(t) + p_{\mathrm{s}}(t) + p_{\mathrm{b}}(t) + p_{\mathrm{h}}(t) = 1$. Then, the posterior probability of each symbol is defined as follows:

- blank probability at frame $t = p_{\mathrm{b}}(t)$
- on probability at frame $t = p_{\mathrm{o}}(t)$
- sil probability at frame $t = p_{\mathrm{s}}(t)$
- Probability of a note pitch symbol v, which is in octave c and has pitch class name of pn, at frame $t = p_{\mathrm{h}}(t) \times p_c(t) \times p_{pn}(t)$

Note that both $p_{\mathrm{b}}(t)$ and $p_{\mathrm{h}}(t)$ are only added to compute the CTC loss. They do not affect the post-processing step of the transcription system.

Finally, we can compute the CTC loss between the posterior probability distributions and the target sequence.

D. Advantages and Disadvantages

Using these two loss functions, we can obtain three different combinations of loss functions, which are only cross-entropy loss, only CTC loss, and the sum of the two losses. In this section, we analyze the advantages and disadvantages of these three loss functions in terms of the two issues discussed in Section I.

1) The Property of Sung Note Onset and Offset Prediction: The first issue is about the property of sung note onset and offset prediction. The onsets and offsets of sung notes are usually smooth [12], making them difficult to be labeled precisely even by expert musicians [6]. However, the presence of onsets and offsets of sung notes is clearer and easier to identify [14]. In other words, while the precise position of the onsets and offsets of sung notes are difficult to label, it is not difficult to identify their rough position. It is reasonable to exploit this observation in model training. Specifically, it is desirable to design a loss function that encourages the model to:

- Predict correct onset, offset, and pitch for each frame.
- If it cannot, then at least predict a *correct note sequence*.

The first objective is stricter than the second objective. By training a model to achieve the first objective perfectly, it can also achieve the second objective. However, considering that the first objective is hard to achieve even by expert musicians, adding the second objective, which is easier to achieve by human perception, can help the model learn the characteristic of sung notes easier.

The first objective can be addressed by using the cross-entropy loss, which requires the prediction of each frame to be the same as the groundtruth label, while the second objective can be easily addressed by the CTC loss, which computes the probability that the model's prediction aligns with the target (groundtruth) sequence.

By combining the advantages of the cross-entropy loss and the CTC loss, the sum of the two losses meets the requirements. On the other hand, using the cross-entropy loss only meets the first objective, while using the CTC loss only meets the second objective. Therefore, the sum of the two loss functions has an advantage in this issue.

2) The Use of Weakly Labeled Data: The second issue is about the use of *weakly labeled* data. We define *strongly labeled* data as pairs of audio and note labels in which the onset and offset times are aligned with the audio. By contrast, for *weakly labeled* data, the onset and offset times do not need to be aligned with the audio. For example, most of the musical scores paired with audio recordings available online are weakly labeled data, since the actual performances usually do not strictly follow the musical scores in terms of the note duration.

Although strong labels contain more information than weak labels, creating strong labels for singing transcription is labor-intensive [6], [14], and difficult to scale up. Therefore, it is desirable to make use of weakly labeled data, which is easier to collect.

However, the frame-level labels are required to compute the cross-entropy loss, which is only available in strongly labeled data. By contrast, since the target sequence of the CTC loss does not include onset and offset timings, no strong labels are required to compute the CTC loss. As defined in Section III-C2, the target sequence $\mathbf{z}$ only contains the information of each note's presence sequentially. Note duration information is not included. Therefore, weakly labeled data can also be involved in model training if the CTC loss is used, which is a key advantage for the combination of losses that include the CTC loss (CTC loss only, and the sum of the two losses).

E. System Overview

In this subsection, we discuss the singing transcription system used in the experiments in detail. The proposed system can be viewed as a combination of Wang and Jang's singing transcription system [14] and the *onsets and frames* architecture [32]. The *onsets and frames* architecture has demonstrated the effectiveness of predicting onset and pitch separately. However, it was originally designed for polyphonic piano transcription, which requires the model capable of predicting multiple note pitch values for each frame, which is different from the task of *monophonic vocal with instruments* singing transcription. Therefore, instead of directly adopting the *onsets and frames* model, we integrate the concept of the *onsets and frames* architecture into a previous singing transcription system [14].

Following the problem formulation in Section III-A, we introduce the three steps of the proposed system sequentially, namely *pre-processing*, *frame-level transcription*, and *post-processing*.

1) Pre-Processing: The flowchart of the proposed transcription system is illustrated in Fig. 1. The grey blocks in the figure are modules that do not contain trainable parameters (e.g. constant-Q transform) or are frozen during model training (e.g. Spleeter). The blue blocks are convolution layers, while the orange blocks are linear layers.

The upper part of Fig. 1 shows the pre-processing step of the system. For the input signal $\mathbf{x}$, we first perform singing voice separation (SVS) on it to extract the vocal $\mathbf{x}_{\mathrm{voc}}$ of the audio. Following previous work [13], [14], we apply Spleeter [16] for SVS. Then, we perform constant-Q transform (CQT) [31] on both $\mathbf{x}$

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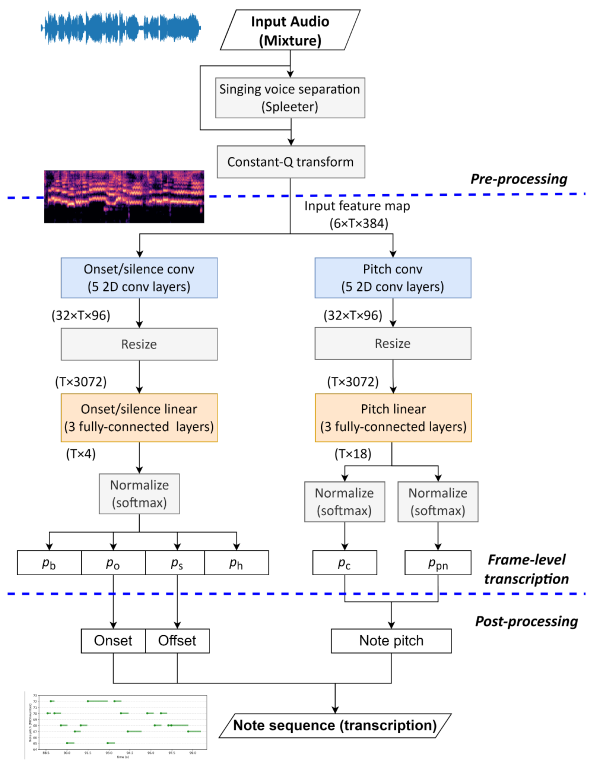

Fig. 1. Flowchart of the proposed singing transcription system, where T represents the total number of frames.

and $\mathbf{x}_{\text{voc}}$ using Librosa package [33] to extract spectral features. The hop length is set to 1024 points (0.023 s), which leads to the frame rate of 43.07 Hz. The number of bins per octave is 48. The total number of frequency bins is $48 \times 8 = 384$, ranging from C1 (32.7 Hz) to C9 (8372 Hz). For both $\mathbf{x}$ and $\mathbf{x}_{\text{voc}}$, we apply three different q values (0.5, 1, and 2) to obtain features with different window sizes, while the hop length remains the same. As a result, $2 \times 3 = 6$ different CQT spectrograms with the same shape of $T \times 384$ are computed, where T is the frame number of the CQT spectrogram. These spectrograms are concatenated to form a $6 \times T \times 384$-d feature map. Note that the mixture $\mathbf{x}$ is also used to form the input. The effectiveness of this setting is verified in Section V-D.

2) Frame-Level Transcription: The $(6 \times T \times 384)$-d feature map is then passed to a convolutional neural network (CNN) for frame-level transcription. As shown in the middle of Fig. 1, the CNN consists of four modules. Two of them are related to onset/offset prediction, while the others are related to pitch prediction. A rectified linear unit (ReLU) function is used as an activation function for each layer in every module except the last layer.

For onset/offset prediction (middle left), the input feature map is passed to the *onset/silence conv* module that consists of 5 2-D convolution layers. The output is then flattened and passed to the *onset/silence linear* module that consists of 3 fully-connected layers, which outputs a 4-d vector for each frame. Finally, a Softmax function is applied to the output vector to produce the predictions of onset probability (p_{o}), silence probability (p_{s}), along with the two dimensions (p_{b} and p_{h}) that are only used for the computation of the CTC loss (as discussed in Section III-C2). Similarly, for pitch prediction (middle right), the input is passed to the *pitch conv* module that consists of 5 2-D convolution layers. The output is then flattened and passed to the *pitch linear* module that consists of 3 fully-connected layers, which outputs a 5-d vector and a 13-d vector. The two vectors are then passed through a Softmax function that produces the predictions of the octave probability (p_c) and pitch class name probability (p_{pn}) for each frame. Then, the results of $p_{\text{o}}(t)$ (onset), $p_{\text{s}}(t)$ (silence), $p_c(t)$ (octave), and $p_{pn}(t)$ (pitch class name) are passed to the post-processing step.

3) Post-Processing: Finally, the post-processing step outputs the note sequence (i.e., the transcription) based on the frame-level prediction. This step can further be divided into three sub-steps, namely onset detection, offset determination, and note pitch determination, as discussed below:

Onset detection: First, we detect notes' onset based on the onset probability predictions. For a frame t, if both:

- $p_{\text{o}}(t) \geq \theta_{\text{o}}$
- $p_{\text{o}}(t) = \max\{p_{\text{o}}(t-l), p_{\text{o}}(t-l+1), \ldots, p_{\text{o}}(t+l)\}$

conditions are satisfied, it is considered as an onset. θ_{o} and l are two hyper-parameters. θ_{o} is the onset threshold, while l affects the length of the local maximum mask. In the experiments, l is empirically fixed at 3 (about 0.07 s long), while the threshold θ_{o} is learned from data, which will be discussed in Section V. The reason for setting l to 3 is that in practice only a few notes have durations less than 0.07 s. Setting l to 3 allows the transcription system to only eliminate notes that are clearly too short.

Offset determination: Then, for each onset, we determine the corresponding offset frame. For an onset located at the frame i, we choose the closest frame t that satisfies:

- $i < t$
- $p_{\text{s}}(t) \geq \theta_{\text{s}}$, or t itself is an onset frame. (in the latter case, the onset of the next note is also the offset of the previous note)

as an offset. A hyper-parameter θ_{s} is applied as the threshold. Similar to θ_{o}, θ_{s} is also learned from data.

Note pitch determination: After the two sub-steps above, we have determined the boundary of each note. To determine the note pitch of a note, we first determine the pitch prediction for each frame within the boundary of the note. For a frame t and any pitch value v, to compute $p_v(t)$, we multiply the corresponding octave probability by the pitch class name probability. For example, MIDI number 69's, or A4's, pitch probability $p_{v=69}(t) = p_{pn=A}(t) \times p_{c=4}(t)$. We choose the pitch value v that maximizes $p_v(t)$ as the pitch prediction. Then, we determine the note pitch of each note as the mode of all frames' pitch prediction within the note boundary, i.e., by majority vote.

As an example, we have plotted the outputs of the singing transcription system on a clip in Fig. 2, along with a comparison of the groundtruth note sequence and the predicted note sequence.

IV. Evaluation Metrics and Dataset Pre-Processing

In this section, we first discuss the evaluation metrics we use to evaluate singing transcription systems, which are the *COn,*

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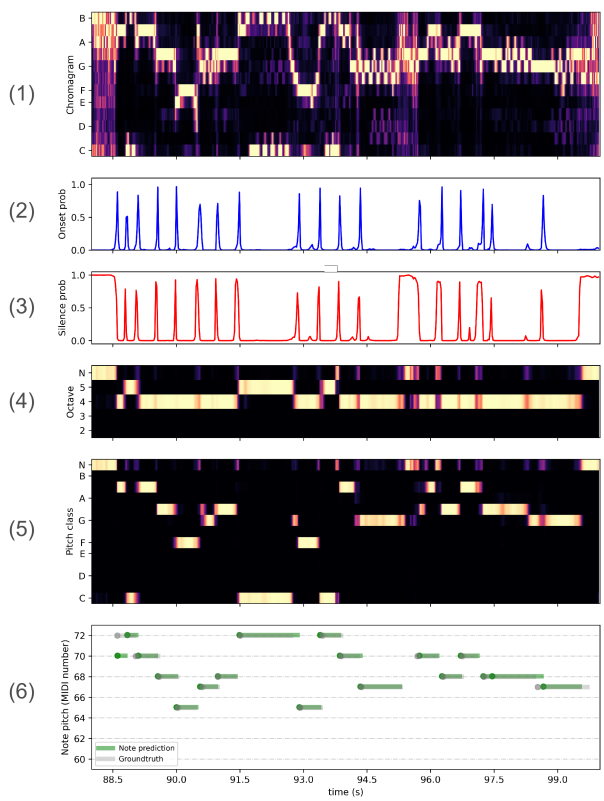

Fig. 2. An example of the outputs of the proposed singing transcription system in different steps. From top to down: (1): The chromagram of the vocals extracted using Spleeter (pre-processing), (2) $\sim$ (5): The prediction of $p_{\mathrm{o}}, p_{\mathrm{s}}, p_c$, and p_{pn} (frame-level transcription) (6): The transcription result (post-processing) and the groundtruth note sequence. In the subplot 4 and 5, p_{N} represents that the frame is a silence frame, so it does not have pitch value.

COnP, COnPOff metrics [6]. Then, we discuss the datasets used in the experiments. Finally, we discuss the proposed *global time shift calibration* method, which is a pre-processing step that deals with the intrinsic *global time shift* between the note labels of different datasets. By calibrating the groundtruth labels of the training dataset to a baseline, a preliminary experiment shows that it alleviates the global time shift issue, especially in the cross-dataset evaluation scenario.

A. Evaluation Metrics

The evaluation metrics adopted in this work are the *COn, COnP, COnPOff* metrics [6], which are widely used in recent papers [8], [10], [12], [14]. COn, COnP, COnPOff metrics propose three different criteria to determine whether a groundtruth note is correctly transcribed. From each criterion, we can determine the number of correctly transcribed notes. Then, by computing the F1-score, the harmonic mean of precision and recall, we can obtain a score between 0 and 1 that represents the performance of a transcription system. Since there are three different criteria, we can obtain three different F1-scores. The three criteria are described as follows [6]:

- *COn:* COn requires that the onset time difference between the groundtruth note and the transcribed note be less than 50 ms.
- *COnP:* COnP requires that the onset time difference between the groundtruth note and the transcribed note be less than 50 ms, and the pitch difference be less than 50 cents (0.5 semitones).
- *COnPOff:* COnPOff requires that the onset time difference between the groundtruth note and the transcribed note be less than 50 ms, the pitch difference be less than 50 cents (0.5 semitones), and the offset difference be less than $max(50\,\mathrm{ms}, 0.2 \times \text{groundtruth note duration})$.

In practice, to avoid the cases that a groundtruth note is correctly transcribed multiple times, or multiple groundtruth notes are transcribed by the same note in the transcription, in the implementation of *mir_eval* package [34], a bipartite matching algorithm is used to compute these metrics, which ensures that one groundtruth note is only transcribed by one note in the transcription, and vice versa. In this paper, we use the implementation of *mir_eval* for model evaluation.

B. Datasets

In this work, we use the training set of the MIR-ST500 dataset [14] to train models, and use the MIR-ST500 dataset's test set and ISMIR2014 dataset [6] to evaluate models. The comparison of the two datasets is shown in Table I.

1) MIR-ST500 Dataset: The MIR-ST500 dataset is a *monophonic vocal with instruments* dataset proposed by Wang and Jang [14]. The dataset consists of 500 pop songs and their corresponding vocal note labels, which are labeled by non-experts. The total audio duration is 34.4 hours, while the total note number is 162,929. The pitch labels of this dataset are *note pitches* which are quantized to integer of semitones. The first 400 songs are used as the training set, while the remaining 100 songs are used as the test set.

2) ISMIR2014 Dataset: The ISMIR2014 dataset is a *monophonic vocal* dataset proposed by Molina et al. [6]. It contains 38 monophonic vocal clips and their corresponding note labels. The total audio duration is 1154 seconds, while the total note number is 2,153. The pitch labels for this dataset are *actual pitches* of floating point numbers, which are directly related to the fundamental frequency (F0) perceived by labelers. In other words, the pitch values are labeled as what the singer sings in the audio. If the singing voice is out of tune, the pitch labels would also follow the singing voice be out of tune. Such a labeling criterion is different from the MIR-ST500 dataset. For the MIR-ST500 dataset, if the singing voice is out of tune, the pitch labels are still the pitches of the notes the singer should sing based on music theory or the context.

Although the ISMIR2014 dataset is not a *monophonic vocal with instruments* dataset, considering that it has been widely used to evaluate transcription systems [8], [10], [12], [14], we also evaluate the singing transcription systems on this dataset, to compare our system with previous work.

C. Global Time Shift Calibration

In this subsection, we introduce the *global time shift calibration* step we have performed as dataset pre-processing. We define the *global time shift* between two datasets as the intrinsic

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TABLE I
COMPARISON BETWEEN DATASETS

Dataset	MIR-ST500 [14]	ISMIR2014 [6]
Audio duration	34.4 hours	1154 seconds
Audio type	*Monophonic vocal with instruments*	*Monophonic vocal*
Number of notes	162,929	2,153
Groundtruth labeler	Non-experts	Experts
Pitch annotation criteria	Note pitch	Actual pitch
Year	2021	2014

TABLE II
THE PERFORMANCE OF THE SAME MODEL TRAINED ON THE SAME DATASET (THE MIR-ST500 TRAINING SET) USING THE SAME LOSS FUNCTION (ONLY THE CROSS-ENTROPY LOSS), BUT USING THE GROUNDTRUTH NOTE LABELS WITH DIFFERENT MANUAL TIME SHIFTS

Training setting	MIR-ST500 test set			ISMIR2014 dataset		
	COnPOff	COnP	COn	COnPOff	COnP	COn
CE (no shift)	**0.5371**	**0.7227**	0.7763	0.5713	0.7072	0.8403
CE (gt shifted)	0.5351	0.7214	**0.7793**	**0.5746**	**0.7487**	**0.9133**

The labels used to train ce (NO SHIFT) are not shifted, while the labels used to train ce (GT SHIFTED) are shifted by +30˜ms

global shift between the note labels of the two datasets. For any two similar sung notes in two different datasets, if the note label of one dataset is usually earlier/later than the note label of the other dataset, then there is a *global time shift* between two datasets.

This issue was observed in a preliminary experiment. Using COn, COnP, COnPOff metrics as evaluation criteria, the preliminary experiment shows that if we manually shift all the groundtruth labels of the MIR-ST500 dataset by $+30\,ms$, and use the shifted note labels of the MIR-ST500 dataset's training set to train a singing transcription model, it can perform better than the model trained using the MIR-ST500 dataset's original labels on the ISMIR2014 dataset.

The experiment results are shown in Table II. *CE (no shift)* is the model trained using the original labels of the MIR-ST500 training set, while *CE (gt shifted)* is the model trained using the shifted labels. Simply applying a $+30\,ms$ time shift on groundtruth labels improves COnP F1-score by more than 4 percentage points, and COn F1-score by more than 7 percentage points on the ISMIR2014 dataset. Note that the performance of the two models on the MIR-ST500 test set (for *CE (gt shifted)*, the groundtruth labels are also shifted by $+30\,ms$) are similar. The largest difference is only 0.3 percentage points on COn F1-score.

These results suggest that the *global time shift* may affect the model performance. Fig. 3 further visualizes its effect by manually shifting the predictions of the models and evaluating the COn F1-scores of the shifted predictions for both the *CE (no shift)* and *CE (gt shifted)* models on the ISMIR2014 dataset. The horizontal axis represents the length of the time shift, while the vertical axis represents the COn F1-score. The blue line indicates the performance of *CE (no shift)*, and the orange dashed line indicates the performance of *CE (gt shifted)*. Both curves have similar shapes (inverse U-shaped) and maximum values (0.9184 and 0.9188). However, the maximum value of the *CE (no shift)* curve occurs when the time shift is $+29.5\,ms$, while the maximum value of the *CE (gt shifted)* curve occurs when the time shift is $-3.0\,ms$. This results in different COn F1-scores for the two models.

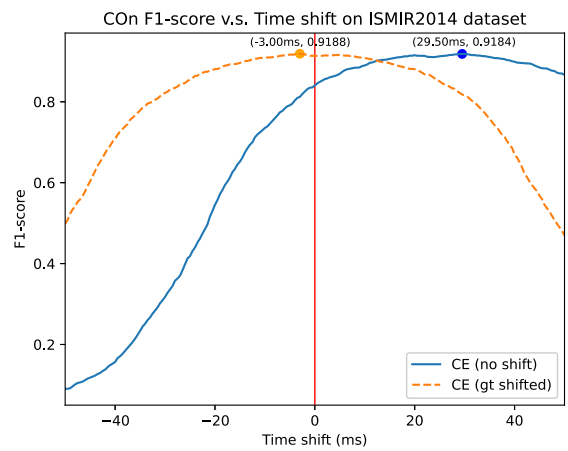

Fig. 3. The COn F1-score versus global time shift for the *CE (no shift)* and *CE (gt shifted)* settings on the ISMIR2014 dataset. The red vertical line indicates that no time shift is applied, which leads to the original COn F1-score.

Furthermore, the approximately $+29.5\,ms$ time shift is only observed between *CE (no shift)*'s prediction on the ISMIR2014 dataset and the groundtruth labels, but not observed between *CE (no shift)*'s prediction on the MIR-ST500 test set and the groundtruth labels. On the other hand, there is no clear time shift between *CE (gt shifted)*'s prediction and the groundtruth labels on both the ISMIR2014 dataset and the shifted MIR-ST500 test set. Based on the above observations, we can conclude that such a global time shift is not caused by the singing transcription model itself, but by the groundtruth labels of the two datasets. Moreover, this issue can be alleviated by simply shifting all the note labels of the MIR-ST500 dataset by $+30\,ms$.

It may be counter-intuitive to think that such a global time shift does exist. However, a global time shift of $30\,ms$ can be caused by several reasons. For example, as mentioned in the

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papers [6], [14], the labelers of the ISMIR2014 and MIR-ST500 datasets were asked to listen to both the audio and the note labels at the same time. This helps labelers create and correct the labels easily. The software used by the labelers may have a slight latency during playback, which eventually results in a global time shift between different datasets. The baseline transcription system used to create the initial note labels may also be a possible cause, since human labelers tend to keep the labels unchanged if the initial labels are not clearly wrong.

Furthermore, the more important issue is that it is difficult to avoid the occurrence of the global time shift, since a slight time shift of $30\,ms$ is too small to be detected by human labelers, which is reflected in the time-consuming labeling process of the datasets [6], [14].

However, in contrast to the low detectability of the global time shift, the results in Table II and Fig. 3 show that a time shift of only $30\,ms$ does affect the model performance in terms of COn F1-score. This is due to the nature of the *COn, COnP, COnPOff* metrics. When there is a $+30\,ms$ global time shift, in practice, the onset timing error that a transcription model is allowed to make become $-20\,ms \sim +80\,ms$. This is more difficult to achieve than the original $-50\,ms \sim +50\,ms$ error tolerance, as a model is barely allowed to make an onset timing error on one side.

To address this issue, we propose a method to automatically calibrate all the note labels of the dataset to a baseline, which does not require any human effort to detect and measure the time shift.

The workflow of the proposed solution is as follows:

- Use a state-of-the-art onset detector to generate onset predictions for a dataset.
- Determine the global time shift s between the onset predictions and the onsets of the groundtruth notes labeled by human labelers.
- Shift all the groundtruth labels by s.

The idea behind this solution is to use an onset detector as a baseline to automatically calibrate the note labels. Since the preliminary experiment results in Table II and Fig. 3 show that among all metrics, COn F1-score is affected the most by the global shift, here we will only consider the COn F1-score. In practice, we use Schlüter and Böck's CNN onset detector [35], a state-of-the-art onset detector, to perform the first step. The threshold for peak picking is set to 0.5, and the size of the local maximum filter is set to $100\,ms$.

For the second step, the target is to compute the global time shift s by:

$$s := \arg\max_{s'} COn(\mathbf{n}_{\text{gt}} + s,' \mathbf{n}_{\text{tr}}) \tag{6}$$

where $\mathbf{n}_{\text{gt}}$ is the onsets of the groundtruth note sequence; $\mathbf{n}_{\text{tr}}$ is the onsets obtained by the onset detector. The $\mathbf{n}_{\text{gt}} + s'$ notation means that all the onsets in $\mathbf{n}_{\text{gt}}$ are shifted by s'. This algorithm finds the best s such that when we shift the onsets in $\mathbf{n}_{\text{gt}}$ by s, the COn F1-score between the shifted groundtruth and the onset predictions is maximized. For the last step, we simply shift the groundtruth labels by the global time shift value s to calibrate the note labels to the baseline.

TABLE III
PARAMETERS OF THE CONVOLUTION LAYERS

Layer\Module	Onset/silence	Pitch
conv1	16, 4, 1	16, 4, 1
conv2	32, 1, 1	32, 1, 1
conv3	32, 1, 1	32, 1, 1
conv4	32, 1, 1	32, 1, 1
conv5	32, 1, 1	32, 1, 1

The 3-tuple in the table stands for {output channel, stride (frequency axis), stride (time axis)}.
All layers have the same kernel size 9 × 9.

By applying the above method, we have found out that the global time shift between the MIR-ST500 training dataset and the baseline is $+30.00\,ms$, which means that all the labels in the MIR-ST500 training dataset should be shifted forward by $30.00\,ms$ to match the prediction of the onset detector. For the ISMIR2014 dataset, the global time shift is $-10.00\,ms$. In other words, the relative global shift between the MIR-ST500 training dataset and the ISMIR2014 dataset is $40.00\,ms$. This result is in line with our preliminary experiment results, as the model trained on the note labels of the MIR-ST500 training set that are manually shifted by $+30\,ms$ performs better on the ISMIR2014 dataset.

Therefore, we propose to shift the note labels of the MIR-ST500 dataset by $+30.00\,ms$ and shift the note labels of the ISMIR2014 dataset by $-10.00\,ms$. However, to ensure a fair comparison with previous work, we choose to not shift the groundtruth of the ISMIR2014 dataset in the experiments in Section V, but only shift the groundtruth of the MIR-ST500 dataset.

V. EXPERIMENT RESULTS

In this section, we first discuss the implementation details of the singing transcription model in Section V-A. We then discuss the experiments we have conducted to verify the effectiveness of the proposed method in both *strongly labeled data* scenario (Section V-B) and the scenario where only a limited number of strong labels and all the weak labels are available (Section V-C). We also compare the performance of our singing transcription system with other state-of-the-art systems in the literature in Section V-B. Finally, an ablation study that examines the effectiveness of using Spleeter [16] in the pre-processing step to extract the singing voice is discussed in Section V-D. The source code of all the experiments is available on GitHub[3].

A. Implementation Details

Tables III and IV show the implementation details of the proposed singing transcription model. For model optimization, we used Adam [36] optimizer to optimize the model with a learning rate of 0.0001. The batch size is set to 1, which means that each batch contains the training data of a whole song. The number of training epochs is set to 100, and early stopping is not used. We stored model parameters (checkpoints) at the end of each

[3][Online]. Available: https://github.com/york135/CTC_CE_for_AST

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TABLE IV
PARAMETERS OF THE LINEAR LAYERS

Layer\Module	Onset/silence	Pitch
fc1	3072, 64	3072, 64
fc2	64, 32	64, 32
fc3	32, 4	32, 5+13[a]

[a] The output of this layer is 18-d, but it is split into a 5-d vector (octave) and a 13-d vector (pitch class name), as discussed in Section III-E.

The 2-tuple in the table stands for {input dimension, output dimension}

epoch, and then perform model selection and hyper-parameter selection according to the following procedures:

- *Select onset threshold:* For each model checkpoint, choose the onset threshold θ_{o} that maximizes the COn F1-score of the training set via an exhaustive search from 41 possible onset thresholds $\{0.1, 0.12, \ldots, 0.88, 0.9\}$. During the search, the offset threshold θ_{s} is fixed at 0.5.
- *Select offset threshold:* For each model checkpoint, choose the offset threshold θ_{s} that maximizes the COnPOff F1-score of the training set via an exhaustive search from 41 possible offset thresholds $\{0.1, 0.12, \ldots, 0.88, 0.9\}$, which is similar to the onset threshold searching. During the search, θ_{o} is fixed at the value determined in the last step.
- *Select checkpoint:* We select the model checkpoint that maximizes the COnPOff F1-score of the training set from all 100 model checkpoints. During the search, both θ_{o} and θ_{s} are fixed at the values determined in the last step.

In a preliminary experiment, we split the dataset to create a validation set, and found that using the training set or validation set for hyper-parameter search leads to similar results. Therefore, for simplicity, instead of separating the validation set from the training set, we simply tune the hyper-parameters on the training set.

B. Strongly Labeled Scenario

This experiment evaluates the effectiveness of the proposed method in the strongly labeled scenario, where all the strong labels in the MIR-ST500's 400-song training set are available. Four different methods are evaluated, including *CE*, *CE smooth*, *CE+CTC*, and *CTC*. The detailed settings are as follows:

- *CE*. Only use the cross-entropy loss for model training. For the groundtruth labels of onset classification, the distance threshold θ_{d} (as discussed in Section III-C1) is set to $0.5\times$hop size. Furthermore, we balance class weights of onset classification to deal with data imbalance. The class weights (32.54 for positive class, 1 for negative class) are derived from the distribution of the training set.
- *CE smooth*. Only use the cross-entropy loss for model training. For the onset cross-entropy loss, θ_{d} is set to 50 ms. This can be viewed as a naive way to deal with the smooth onset property of sung notes, as discussed in Section III-D. For the weights of onset classification, to encourage the model to output the same onsets to the groundtruth, we assign different weights to each onset frame according to its distance from the nearest onset. Suppose the distance between a positive onset frame t and its closest onset is d_t, the weight is set to $w \times \frac{50ms - d_t}{50\,ms}$, where w is a hyper-parameter set to 20 empirically, which makes the total weights of positive onset frames roughly the same as negative onset frames.
- *CE+CTC*. Use the sum of the cross-entropy loss and the CTC loss for model training. Since the scales of the two loss functions are similar (which is observed in a preliminary experiment), we simply assign equal weights to the two losses. The settings for the cross-entropy loss are the same as *CE smooth*.
  During model training, to make the model easier to converge, in the first 5 epochs, only the cross-entropy loss is used. Then, both loss functions are used to train the model in the remaining 95 epochs.
- *CTC*. Only use the CTC loss for model training.

To verify the effectiveness of the proposed method, the experiments of the four methods are repeated 5 times. The 95% confidence intervals are reported, and independent t-tests are also conducted to test the statistical significance of different methods. The results are shown in Table V. The performance of the model on the MIR-ST500 test set and ISMIR2014 dataset are reported. Regardless of the evaluation criteria and the dataset used for evaluation, *CE+CTC* achieves the best performance among the four methods.

On the MIR-ST500 test set, *CE+CTC* outperforms all other methods by at least 2 percentage points on all metrics. The improvements are all statistically significant ($p < 10^{-5}$ on all metrics). These results demonstrate the positive effect of using both the cross-entropy loss and the CTC loss for model optimization. As for other methods, the performance of *CE* and *CE smooth* are similar, suggesting that simply smoothing the onset labels does not lead to improvement. The reason is that while smoothing the onset labels does address the smooth nature of the onsets of sung notes, it does not take into account the observation that *the presence of sung notes' onsets are easier to detect*, which is the first issue discussed in Section III-D. By contrast, the *CE+CTC* method addresses the issue by using the combination of the cross-entropy loss and the CTC loss for model optimization, resulting in improved performance.

For *CTC*, its performance falls far behind other methods. It also has a larger 95% confidence interval than the other methods. By further examining the models' predictions, we found that all the models trained with *CTC* method do output meaningful notes. However, three of the five models predict notes that globally deviate from the groundtruth notes by more than 100 ms (i.e., these models introduce a large *global time shift* by themselves wrongly), which is at least two times larger than the 50 ms onset tolerance, resulting in significantly lower F1-scores. This can be verified by setting the onset tolerance to 150 ms, which leads to 0.6004 COn F1-score and 0.5211 COnP F1-score on the MIR-ST500 test set. Both of them are about two times higher than the corresponding F1-scores under the 50 ms onset tolerance condition (0.3054 for COn, 0.2555 for COnP).

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TABLE V
PERFORMANCE (F1-SCORES) OF DIFFERENT METHODS IN THE *STRONGLY LABELED SCENARIO*

Method	MIR-ST500 test set			ISMIR2014 dataset		
	COnPOff	COnP	COn	COnPOff	COnP	COn
CE	0.5304 ± 0.0033	0.7196 ± 0.0016	0.7747 ± 0.0034	0.5692 ± 0.0090	0.7543 ± 0.0101	0.9101 ± 0.0050
CE smooth	0.5336 ± 0.0055	0.7146 ± 0.0034	0.7700 ± 0.0020	0.6026 ± 0.0177	0.7614 ± 0.0176	0.9111 ± 0.0079
CE+CTC	$\mathbf{0.5739 \pm 0.0045}$	$\mathbf{0.7437 \pm 0.0037}$	$\mathbf{0.7962 \pm 0.0030}$	$\mathbf{0.6360 \pm 0.0156}$	$\mathbf{0.7673 \pm 0.0152}$	$\mathbf{0.9301 \pm 0.0046}$
CTC	0.1071 ± 0.1639	0.2555 ± 0.2540	0.3054 ± 0.2834	0.1290 ± 0.1821	0.2724 ± 0.2602	0.3562 ± 0.3450

The 95% confidence intervals are also reported. for each metric, the best result among all methods is shown in bold.

The result that *CTC* does not lead to acceptable model performance is different from previous papers, in which using only the CTC loss still leads to comparable model performance in other MIR tasks [18], [25], [26], [27], [28], [30]. As discussed in Section III-D, we suggest that the main difference is that the task of singing transcription requires the onset and offset timings of each note to be predicted precisely (only 50 ms error tolerance). This specific requirement is difficult to achieve using only the CTC loss, since the CTC loss does not encourage the model to predict the correct onset and offset times of sung notes. Therefore, using only the CTC loss does not always lead to high F1-scores in the task of note-level singing transcription.

On the ISMIR2014 dataset, *CE+CTC* significantly outperforms all other methods on COn and COnPOff F1-scores ($p < 0.003$). However, the improvement is not statistically significant on COnP F1-score comparing to *CE smooth* ($p = 0.25$). The reason may be that the pitch labels in the ISMIR2014 dataset are actual pitches, but not note pitches. In the case when the singing voice is out of tune, the actual pitch and note pitch labels will be different. Therefore, a model trained on the MIR-ST500 training set (pitch labels are note pitch) may not necessarily predict the actual pitch well. However, in general, the *CE+CTC* method still outperforms other methods on the ISMIR2014 dataset, which verifies that using the combination of the cross-entropy loss and the CTC loss is beneficial for training a singing transcription model.

Comparison with previous work: Furthermore, we compare the model trained using the proposed *CE+CTC* method with previous work in the literature on various datasets. We used both the MIR-ST500 test set and the ISMIR2014 dataset to evaluate the performance of the singing transcription systems.

For a fair comparison, we only consider note-level transcription systems that predict a note sequence, instead of a musical score with quantized notes (such as [37], [38]), which are related to *notation-level* singing transcription. As a result, we include Omnizart[4] [13], Wang and Jang [14], Gómez and Bonada [5], Tony [10], Yang et al. [8], and Fu and Su [12] in the comparison. Gómez and Bonada [5] is a rule-based singing transcription system; Tony [10] and Yang et al. [8] are based on hidden Markov model (HMM); Omnizart [13], Wang and Jang [14] and Fu and Su [12] are based on convolutional neural network (CNN). More details are discussed in Section II.

Among these systems, only Omnizart [13] and Wang and Jang [14] are *monophonic vocal with instruments* singing transcription systems. All the others are *monophonic vocal* singing transcription systems. Therefore, for the MIR-ST500 test set (a *monophonic vocal with instruments* dataset), only the first two systems are used for comparison, while for the ISMIR2014 dataset (a *monophonic vocal* dataset), all the systems are included for comparison.

[4]We ran Omnizart's *"omnizart vocal transcribe"* command to perform singing transcription.

TABLE VI
COMPARISON OF F1-SCORES OF THE PROPOSED *CE+CTC* METHOD WITH PREVIOUS *MONOPHONIC VOCAL WITH INSTRUMENTS* SINGING TRANSCRIPTION SYSTEMS ON THE MIR-ST500 TEST SET

Method \ Metric	COnPOff	COnP	COn
Omnizart [13]	0.2115	0.3831	0.5047
Wang & Jang [14]	0.4580	0.6664	0.7543
CE+CTC	**0.5739**	**0.7437**	**0.7962**

For each metric, the best result is shown in bold.

TABLE VII
COMPARISON OF F1-SCORE OF THE PROPOSED *CE+CTC* METHOD WITH PREVIOUS WORK ON THE ISMIR2014 DATASET, A MONOPHONIC DATASET

Method \ Metric	COnPOff	COnP	COn
Gómez & Bonada [5]	0.39	0.55	0.66
Tony [10]	0.50	0.68	0.73
Yang et al. [8]	0.421	NA[a]	NA[a]
Fu & Su [12]	0.594	NA[a]	0.786
Omnizart [13]	0.5018	0.6330	0.7755
Wang & Jang [14]	0.4955	0.6363	0.7916
CE+CTC	**0.6360**	**0.7673**	**0.9301**

[a] "NA" means that this metric is not reported in their paper. For each metric, the best result is shown in bold.

Table VI shows the comparison of Omnizart [13], Wang and Jang [14], and *CE+CTC* on the MIR-ST500 test set. The bold text indicates the best performance among all systems on a metric. Our model outperforms Wang and Jang [14] by more than 4 percentage points on all metrics and Omnizart [13] by more than 29 percentage points on all metrics, showing that the proposed transcription system outperforms the state-of-the-art on *monophonic vocal with instruments* singing transcription.

For the ISMIR2014 dataset, the performance of various systems is shown in Table VII. The proposed method outperforms all previous works by at least 8 percentage points on all metrics. The results show that the proposed model also achieves state-of-the-art in transcribing *monophonic vocal* music.

C. Weakly Labeled Scenario

This experiment evaluates the effectiveness of the proposed method when only a small amount of strongly labeled data and a relatively large amount of weakly labeled data are available. We abbreviate this scenario as *weakly labeled scenario* (as opposed to *strongly labeled scenario* in the last subsection). In practice,

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TABLE VIII
PERFORMANCE (F1-SCORE) OF DIFFERENT METHODS IN THE *WEAKLY LABELED SCENARIO*

Method	MIR-ST500 test set			ISMIR2014 dataset		
	COnPOff	COnP	COn	COnPOff	COnP	COn
CE10	0.2694 ± 0.0279	0.4442 ± 0.0476	0.5723 ± 0.0467	0.3223 ± 0.0296	0.5085 ± 0.0268	0.6971 ± 0.0485
CE10+CTC400	0.4860 ± 0.0133	0.6712 ± 0.0167	0.7513 ± 0.0113	0.5284 ± 0.0126	0.7114 ± 0.0324	0.8816 ± 0.0258
CE100	0.4158 ± 0.0082	0.6221 ± 0.0065	0.7024 ± 0.0096	0.4520 ± 0.0525	0.6538 ± 0.0167	0.8230 ± 0.0053
CE100+CTC400	$\mathbf{0.5388 \pm 0.0106}$	0.7124 ± 0.0094	$\mathbf{0.7765 \pm 0.0083}$	$\mathbf{0.6127 \pm 0.0509}$	0.7476 ± 0.0375	$\mathbf{0.9168 \pm 0.0248}$
CE400	0.5304 ± 0.0033	$\mathbf{0.7196 \pm 0.0016}$	0.7747 ± 0.0034	0.5692 ± 0.0090	$\mathbf{0.7543 \pm 0.0101}$	0.9101 ± 0.0050

The 95% confidence intervals are also reported. for each metric, the best result among all settings is shown in bold. for comparison, the results of *CE* in strongly labeled scenario are also reported (DENOTED AS *CE400*).

we still use the MIR-ST500 training set for model training, but only allow the model to access a limited amount of strong labels. We test the settings where only 10 or 100 strongly labeled songs are available. For all the other songs in the MIR-ST500 training set, we only provide the model with weak labels, i.e., note sequences without exact onset and offset times of notes.

Four different settings are evaluated, including *CE10+CTC400*, *CE100+CTC400*, *CE10* and *CE100*. The number after *CE* indicates the amount of strongly labeled data used to compute the cross-entropy loss, while the number after *CTC* indicates the amount of weakly labeled data used to compute the CTC loss. For example, *CE10+CTC400* means that only 10 strongly labeled songs are used to compute the cross-entropy loss, and all the 400 songs are used to compute the CTC loss. For *CE10* and *CE100*, the CTC loss is not used. They serve as baselines of only using the cross-entropy loss.

Since the number of available strongly labeled data is less than the weakly labeled data, for *CE10+CTC400* and *CE100+CTC400*, in each iteration of model training, we randomly sample one song from the available strongly labeled data with replacement to compute the cross-entropy loss. That is, the same data may appear multiple times in each epoch.

To verify the effectiveness of the proposed methods, the experiments of the four settings are repeated 3 times. The 95% confidence intervals are reported, and independent t-tests are also conducted. The results are shown in Table VIII. Both *CE100+CTC400* and *CE10+CTC400* outperform their corresponding cross-entropy baselines (*CE100* and *CE10*) significantly ($p < 0.002$). Furthermore, even *CE10+CTC400* outperforms *CE100* significantly ($p < 0.01$), demonstrating the effectiveness of the proposed *CE+CTC* method in weakly labeled scenario.

In addition, we have also put the results of *CE* in the strongly labeled scenario in the last row of Table VIII. Since in the strongly labeled scenario, all 400 strongly labeled data are used for training, this setting is denoted as *CE400* here. Results show that the performance of *CE100+CTC400* and *CE400* are similar. That is to say, using the proposed *CE+CTC* method, only 100 strongly labeled songs and 400 weakly labeled songs are sufficient to train a model that performs similarly to the traditional *CE* method with 400 strongly labeled songs. These results are promising, showing the potential of using a large-scale weakly labeled dataset for training singing transcription models.

Furthermore, we have conducted another experiment that changes the number of available weakly labeled data. The number of strongly labeled data is fixed at 10. The number of weakly labeled data includes 0, 10, 50, 100, and 400. Therefore, the five settings can be denoted as *CE10*, *CE10+CTC10*, *CE10+CTC50*, *CE10+CTC100* and *CE10+CTC400*. Fig. 4 shows the performance curve of F1-scores versus the number of weakly labeled data used for model training. The models are evaluated on the MIR-ST500 test set. The results show that increasing the number of weakly labeled data does lead to performance improvement, even if the number of weakly labeled data has been significantly larger than strongly labeled data, which is usually the practical case because weakly labeled data is easier to collect than strongly labeled data. For example, the improvement of *CE10+CTC400* over *CE10+CTC100* is $2 \sim 3$ percentage points.

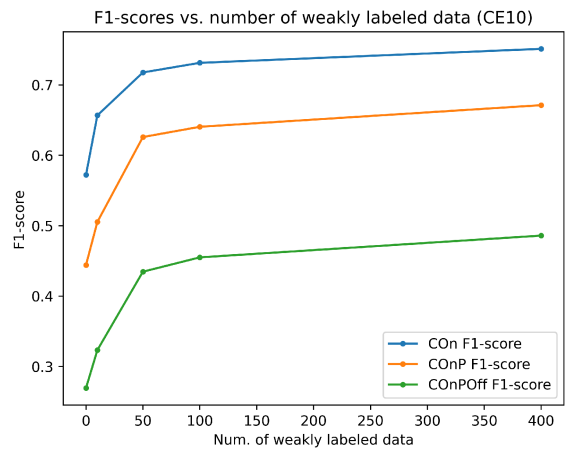

Fig. 4. The performance of the model trained on 10 strongly labeled songs and various numbers of weakly labeled songs (0, 10, 50, 100, and 400 songs), on the MIR-ST500 test set.

D. Ablation Study

The last experiment is an ablation study, in which we examine the effectiveness of using the open-source SVS model Spleeter [16] for pre-processing. In this experiment, we test the settings of *CE+CTC (mix)* and *CE+CTC (voc)*. Both settings use the same training method and training data as *CE+CTC*. The only difference is that *CE+CTC (mix)* only takes the CQT spectrograms of the mixture (original audio) as input, while *CE+CTC (voc)* only takes the spectrograms of the vocals extracted by Spleeter as input.

Both the experiments of *CE+CTC (mix)* and *CE+CTC (voc)* are repeated 3 times. The results are shown in Table IX. For

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TABLE IX
PERFORMANCE (F1-SCORES) COMPARISON OF *CE+CTC*, *CE+CTC (MIX)* AND *CE+CTC (VOC)*

Method	MIR-ST500 test set			ISMIR2014 dataset		
	COnPOff	COnP	COn	COnPOff	COnP	COn
CE+CTC	$\mathbf{0.5739 \pm 0.0045}$	$\mathbf{0.7437 \pm 0.0037}$	$\mathbf{0.7962 \pm 0.0030}$	0.6360 ± 0.0156	0.7673 ± 0.0152	0.9301 ± 0.0046
CE+CTC (mix)	0.5451 ± 0.0100	0.7243 ± 0.0074	0.7776 ± 0.0065	0.6248 ± 0.0293	$\mathbf{0.7769 \pm 0.0137}$	0.9223 ± 0.0146
CE+CTC (voc)	0.5597 ± 0.0102	0.7301 ± 0.0067	0.7833 ± 0.0067	$\mathbf{0.6380 \pm 0.0173}$	0.7723 ± 0.0041	$\mathbf{0.9307 \pm 0.0050}$

The 95% confidence intervals are also reported. for each metric, the best result among all settings is shown in bold.

comparison, we also list the results of *CE+CTC* in the table. According to the results, *CE+CTC* outperforms both *CE+CTC (mix)* and *CE+CTC (voc)* on all metrics on the MIR-ST500 test set significantly ($p < 0.005$). These results demonstrate the effectiveness of using an SVS model to extract vocals from the mixture. However, the fact that *CE+CTC* also outperforms *CE+CTC (voc)* significantly on the MIR-ST500 test set suggests that only using the vocals extracted by Spleeter is still not optimal. The reason may be that Spleeter does not extract vocals perfectly. Therefore, it is desirable to add the mixture as input to avoid error propagation.

On the other hand, the F1-scores of models trained using the three settings on the ISMIR2014 dataset are similar. There is no statistical significance observed between any two of these settings on any of the metrics (all p values are larger than 0.05), which is reasonable since the audio of the ISMIR2014 dataset does not contain any instruments.

VI. CONCLUSIONS AND FUTURE WORK

We have proposed a novel method for training a singing transcription model using the combination of the CTC loss and the cross-entropy loss. The use of the CTC loss encourages the model to predict the correct *note sequence* rather than only the frame-level onset/silence/pitch labels, which eventually leads to improved model performance as shown in the experiment results. Furthermore, applying the CTC loss allows *weakly labeled data* that only contains unaligned note sequence labels to be used for model training. Moreover, we have discussed the issue of global time shift between note labels from different singing transcription datasets and introduced a method to resolve this issue.

A possible direction of future work is to apply the proposed *CE+CTC* method to specific transcription tasks (e.g. traditional or local musical instruments) for which a large-scale densely labeled dataset does not exist. In addition, the *CE+CTC* method can be extended to polyphonic transcription. For this task, the MCTC loss [30], an extension of the CTC loss, could be used to compute the CTC loss when multiple labels are present simultaneously. As future work, we would like to verify whether this approach could also lead to better performance on polyphonic singing transcription.

ACKNOWLEDGMENT

The authors would like to thank Chung-Che Wang, Tzu-Chun Yeh, Quinn McGarry, and Li Su for advising on the manuscript.

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